

Lab 10

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Part 1 – CRUD Without Django Forms

Task 1: List Books

URL: `/books/lab10_part1/listbooks`

urls.py

```
path('lab10_part1/listbooks', views.lab10_part1_listbooks,
     name="books.lab10_part1_listbooks"),
```

views.py

```
def lab10_part1_listbooks(request):
    books = Book.objects.all()
    return render(request, 'bookmodule/lab10_part1_listbooks.html',
                  {'books': books})
```

lab10_part1_listbooks.html

```
<a href="{% url 'books.lab10_part1_addbook' %}">+ Add New Book</a>
<table>
  {% for book in books %}
  <tr>
    <td>{{ book.id }}</td>
    <td>{{ book.title }}</td>
    <td>{{ book.price }}</td>
    <td>{{ book.quantity }}</td>
    <td>{{ book.rating }}</td>
    <td>
      <a href="{% url 'books.lab10_part1_editbook' book.id %}">Edit</a>
      <a href="{% url 'books.lab10_part1_deletebook' book.id %}">Delete</a>
    </td>
  </tr>
  {% endfor %}
</table>
```

Screenshot

[←](#) [→](#) [↻](#) http://127.0.0.1:8000/books/lab10_part1/listbooks [🔍](#) [📄](#) [📁](#) [⋮](#)

[Home](#) [List Books](#) [About](#)

Book List

+ Add New Book

ID	Title	Price (\$)	Quantity	Rating	Actions
1	Continuous Delivery	120.0	1	1	Edit Delete
2	Reversing: Secrets of Reverse Engineer	97.0	1	1	Edit Delete
3	The Hundred-Page Machine Learning Book	100.0	1	1	Edit Delete
4	The Great Gatsby	454.15	1	1	Edit Delete
5	Harry Potter and the Sorcerer's Stone	102.9	1	1	Edit Delete
6	Pride and Prejudice	279.1	1	1	Edit Delete

Task 2: Add Book

URL: `/books/lab10_part1/addbook`

urls.py

```
path('lab10_part1/addbook', views.lab10_part1_addbook,
      name="books.lab10_part1_addbook"),
```

views.py

```
def lab10_part1_addbook(request):
    if request.method == 'POST':
        title = request.POST.get('title')
        price = request.POST.get('price')
```

```
quantity = request.POST.get('quantity')
pubdate = request.POST.get('pubdate')
rating = request.POST.get('rating')
author_id = request.POST.get('author_id')
authorObj = Author.objects.get(id=author_id)
obj = Book(title=title, price=float(price), quantity=int(quantity),
           pubdate=pubdate or None, rating=int(rating))
obj.save()
obj.authors.add(authorObj)
return redirect('books.lab10_part1_listbooks')
authors = Author.objects.all()
return render(request, 'bookmodule/lab10_part1_addbook.html',
              {'authors': authors})
```

lab10_part1_addbook.html

```
<form method="post" action="{% url 'books.lab10_part1_addbook' %}">
  {% csrf_token %}
  <input type="text" name="title" required>
  <input type="number" name="price" step="0.01" value="0.0">
  <input type="number" name="quantity" value="1">
  <input type="date" name="pubdate">
  <input type="number" name="rating" min="1" max="5" value="1">
  <select name="author_id" required>
    {% for author in authors %}
      <option value="{{ author.id }}">{{ author.name }}</option>
    {% endfor %}
  </select>
  <button type="submit">Save</button>
</form>
```

Screenshot

Home List Books About

Add New Book

Title

Price

Quantity

Publication Date

Rating (1–5)

Author

Save Cancel

Task 3: Edit Book

URL: `/books/lab10_part1/editbook/<id>`

urls.py

```
path('lab10_part1/editbook/<int:bookId>', views.lab10_part1_editbook,
name="books.lab10_part1_editbook"),
```

views.py

```
def lab10_part1_editbook(request, bookId):
    book = Book.objects.get(id=bookId)
    if request.method == 'POST':
        book.title = request.POST.get('title')
        book.price = float(request.POST.get('price') or 0)
        book.quantity = int(request.POST.get('quantity') or 1)
        book.pubdate = request.POST.get('pubdate') or None
        book.rating = int(request.POST.get('rating') or 1)
        author_id = request.POST.get('author_id')
```

```
authorObj = Author.objects.get(id=author_id)
book.save()
book.authors.set([authorObj])
return redirect('books.lab10_part1_listbooks')
authors = Author.objects.all()
return render(request, 'bookmodule/lab10_part1_editbook.html', {'book':
book, 'authors': authors})
```

lab10_part1_editbook.html

```
<form method="post" action="{% url 'books.lab10_part1_editbook' book.id
%}">
  {% csrf_token %}
  <input type="text" name="title" value="{{ book.title }}">
  <input type="number" name="price" value="{{ book.price }}">
  <input type="number" name="quantity" value="{{ book.quantity }}">
  <input type="date" name="pubdate" value="{{ book.pubdate|date:'Y-m-d'
  }}">
  <input type="number" name="rating" value="{{ book.rating }}">
  <select name="author_id" required>
    {% for author in authors %}
      <option value="{{ author.id }}" {% if author in book.authors.all
%}selected{% endif %}>{{ author.name }}</option>
    {% endfor %}
  </select>
  <button type="submit">Save</button>
</form>
```

Screenshot

[←](#) [→](#) [↺](#) http://127.0.0.1:8000/books/lab10_part1/editbook/1 [🔍](#) [📄](#) [📁](#) [📧](#) [...](#)

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Edit Book (ID: 1)

Title

Price

Quantity

Publication Date
 

Rating (1–5)

Author
 

Task 4: Delete Book

URL: `/books/lab10_part1/deletebook/<id>`

urls.py

```
path('lab10_part1/deletebook/<int:bookId>', views.lab10_part1_deletebook,
name="books.lab10_part1_deletebook"),
```

views.py

```
def lab10_part1_deletebook(request, bookId):
    Book.objects.filter(id=bookId).delete()
    return redirect('books.lab10_part1_listbooks')
```

No template, clicking Delete in the list triggers this URL directly and redirects back.

Screenshot

The screenshot displays a Django application interface. On the left, the `forms.py` file is open in a code editor, showing the `BookForm` class and its `Meta` class. The `BookForm` class uses `ModelChoiceField` for the `author` field and `Select` widget for the `pubdate` field. The `Meta` class defines the `fields` and `widgets` for the form. On the right, the `Book List` page is shown, featuring a table with columns: ID, Title, Price (\$), Quantity, Rating, and Actions. The table lists seven books, each with an 'Edit' and 'Delete' button. Below the table, there is a '+ Add New Book' button. The bottom of the screenshot shows the terminal output, indicating the application is running successfully.

```
177 ## Task 1: Reimplement CRUD using Django forms
184 from .models import Book, Author
185
186 class BookForm(forms.ModelForm):
187     author = forms.ModelChoiceField(
188         empty_label=None,
189         queryset=Author.objects.all(),
190         required=True,
191         label="Author",
192         widget=forms.Select(attrs={
193             'class': 'mycssclass',
194             'id': 'jsID2'
195         })
196     )
197
198 class Meta:
199     model = Book
200     fields = ('title', 'price', 'quantity',
201               'pubdate', 'rating')
202     widgets = {
203         'pubdate': forms.DateInput(attrs=
204             {'type': 'date'}),
205     }
206
207 **urls.py**
208 ``python
209 path('lab9_part2/listbooks', views.
210     lab9_part2_listbooks, name="books.
211     lab9_part2_listbooks"),
212 path('lab9_part2/addbook', views.
213     lab9_part2_addbook, name="books.
214     lab9_part2_addbook"),
215 path('lab9_part2/editbook/<int:bookId>', views.
216     lab9_part2_editbook, name="books.
217     lab9_part2_editbook"),
218 path('lab9_part2/deletebook/<int:bookId>', views.
219     lab9_part2_deletebook, name="books.
220     lab9_part2_deletebook")
```

ID	Title	Price (\$)	Quantity	Rating	Actions
2	Reversing: Secrets of Reverse Engineer	97.0	1	1	Edit Delete
3	The Hundred-Page Machine Learning Book	100.0	1	1	Edit Delete
4	The Great Gatsby	454.15	1	1	Edit Delete
5	Harry Potter and the Sorcerer's Stone	102.9	1	1	Edit Delete
6	Pride and Prejudice	279.1	1	1	Edit Delete
7	1984	115.68	1	1	Edit Delete

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python - libraryproject

```
(djangoenv) Rudi libraryproject > python3 manage.py runserver
/Users/rudi/Library/Mobile Documents/com~apple~CloudDocs/College/Web/Web_Tech/DjangoProjects/djangoenv/lib/python3.13/site-packages/django/db/models/fields/__init__.py:16
70: RuntimeWarning: DateTimeField Book.pubdate received a naive datetime (2026-05-07 00:00:00) while time zone support is active.
warnings.warn(
[04/May/2026 06:58:08] "POST /books/lab10_part1/addbook HTTP/1.1" 302 0
[04/May/2026 06:58:08] "GET /books/lab10_part1/listbooks HTTP/1.1" 200 27316
[04/May/2026 06:58:34] "GET /books/lab10_part1/addbook HTTP/1.1" 200 3039
[04/May/2026 06:58:59] "GET /books/lab10_part1/listbooks HTTP/1.1" 200 27316
[04/May/2026 07:00:11] "GET /books/lab10_part1/editbook/1 HTTP/1.1" 200 3099
[04/May/2026 07:00:50] "GET /books/lab10_part1/deletebook/1 HTTP/1.1" 302 0
[04/May/2026 07:00:50] "GET /books/lab10_part1/listbooks HTTP/1.1" 200 26930
```

Part 2 – CRUD With Django Forms

Task 1: Reimplement CRUD using Django forms

URLs: `/books/lab10_part2/listbooks`, `/addbook`, `/editbook/<id>`, `/deletebook/<id>`

forms.py

```
from django import forms
from .models import Book, Author

class BookForm(forms.ModelForm):
    author = forms.ModelChoiceField(
        empty_label=None,
        queryset=Author.objects.all(),
        required=True,
        label="Author",
        widget=forms.Select(attrs={
            'class': 'mycssclass',
            'id': 'jsID2'
        })
    )

    )
```

```
class Meta:
    model = Book
    fields = ['title', 'price', 'quantity', 'pubdate', 'rating']
    widgets = {
        'pubdate': forms.DateInput(attrs={'type': 'date'}),
    }
```

urls.py

```
path('lab9_part2/listbooks', views.lab9_part2_listbooks,
name="books.lab9_part2_listbooks"),
path('lab9_part2/addbook', views.lab9_part2_addbook,
name="books.lab9_part2_addbook"),
path('lab9_part2/editbook/<int:bookId>', views.lab9_part2_editbook,
name="books.lab9_part2_editbook"),
path('lab9_part2/deletebook/<int:bookId>', views.lab9_part2_deletebook, name=
"books.lab9_part2_deletebook"),
```

views.py

```
def lab9_part2_listbooks(request):
    books = Book.objects.all()
    return render(request, 'bookmodule/lab9_part2_listbooks.html',
{'books': books})

def lab9_part2_addbook(request):
    obj = None
    authors = Author.objects.all()
    if request.method == 'POST':
        form = BookForm(request.POST)
        if form.is_valid():
            obj = form.save()
            obj.authors.add(form.cleaned_data['author'])
            return redirect('books.lab9_part2_listbooks')
    else:
        form = BookForm(None)
    return render(request, 'bookmodule/lab9_part2_addbook.html',
{'authors': authors, 'form': form})

def lab9_part2_editbook(request, bookId):
    book = Book.objects.get(id=bookId)
    authors = Author.objects.all()
    if request.method == 'POST':
        form = BookForm(request.POST, instance=book)
        if form.is_valid():
            obj = form.save()
            obj.authors.set([form.cleaned_data['author']])
            return redirect('books.lab9_part2_listbooks')
    else:
```



```
        form = BookForm(instance=book)
    return render(request, 'bookmodule/lab9_part2_editbook.html',
{'authors': authors, 'form': form, 'book': book})

def lab9_part2_deletebook(request, bookId):
    Book.objects.filter(id=bookId).delete()
    return redirect('books.lab9_part2_listbooks')
```

lab9_part2_addbook.html / lab9_part2_editbook.html (key section)

```
<form method="post">
    {% csrf_token %}
    {{ form.as_p }}
    <button type="submit">Save</button>
</form>
```

`{{ form.as_p }}` renders each field with its label and displays validation errors automatically.

Screenshots

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http://127.0.0.1:8000/books/lab9_part2/listbooks

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Home

List Books

About

Book List (Part 2)

+ Add New Book

ID	Title	Price (\$)	Quantity	Rating	Actions
2	Reversing: Secrets of Reverse Engineer	97.0	1	1	Edit Delete
3	The Hundred-Page Machine Learning Book	100.0	1	1	Edit Delete
4	The Great Gatsby	454.15	1	1	Edit Delete
5	Harry Potter and the Sorcerer's Stone	102.9	1	1	Edit Delete
6	Pride and Prejudice	279.1	1	1	Edit Delete
7	1984	115.68	1	1	Edit Delete

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Home

List Books

About

Add New Book (Part 2)

Title:

Price:

0.0

Quantity:

1

Pubdate:

dd/mm/yyyy

📅

Rating:

1

Author:

Andrew Hunt

▼

Save

Cancel